Unit 6 · Session 2 · Tue, Dec 1, 2026

Genitourinary & Reproductive Pharmacology — Session 2

4 parts · every fact cited to its slide or textbook page. About 120 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

55 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

10 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

30 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Bladder map: **M3** contracts the detrusor (void); **alpha1** closes the neck and **beta3** relaxes the detrusor (store); somatic nicotinic holds the external sphincter. S4; S7; S9; SG U6-R11

2. **Antimuscarinics** block M3 — dry mouth, constipation, confusion; **oxybutynin** has the highest CNS burden → older adults get **mirabegron** (beta3; may raise BP). S13; S15; S17; SG U6-R12; SG U6-R13

3. **Alpha1 blockers** relax the outlet in days (orthostasis, first dose, floppy iris); tamsulosin/silodosin are uroselective. S21; SG U6-R14

4. **5-ARIs** shrink the prostate over 3–6 months and **halve PSA**; often combined with an alpha1 blocker. S23; S24; SG U6-R15

5. **PDE5 inhibitors** keep cGMP high; **any nitrate = absolute contraindication** (and caution with alpha1 blockers). S26; S29; SG U6-R16

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO6.1 Identify major pharmacologic classes used in the management of genitourinary and reproductive conditions and describe their mechanisms of action.

Part 1 · Autonomic Control of the Lower Urinary Tract
Part 2 · Overactive Bladder Drugs
Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors

IO6.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat genitourinary and reproductive conditions.

Part 2 · Overactive Bladder Drugs
Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors

IO6.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 2 · Overactive Bladder Drugs
Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors
Part 4 · Putting It All Together: GU Toolkit

IO6.4 Select appropriate pharmacologic therapy for common genitourinary and reproductive conditions based on clinical presentation and patient-specific factors.

Part 2 · Overactive Bladder Drugs
Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors
Part 4 · Putting It All Together: GU Toolkit

IO6.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in genitourinary and reproductive conditions.

Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors

IO6.6 Explain important medication-use and safety information to patients receiving genitourinary and reproductive pharmacotherapy.

Part 2 · Overactive Bladder Drugs
Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors
Part 4 · Putting It All Together: GU Toolkit

SO1 Describe the autonomic and somatic innervation of the bladder and lower urinary tract, and the receptor subtypes governing storage versus voiding.

Part 1 · Autonomic Control of the Lower Urinary Tract

SO2 Describe the mechanism, indications, and adverse effects of antimuscarinic agents and beta-3 agonists in overactive bladder and urinary incontinence.

Part 2 · Overactive Bladder Drugs

SO3 Describe the mechanism and adverse effects of alpha1 blockers and 5-alpha-reductase inhibitors in benign prostatic hyperplasia (BPH).

Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors

SO4 Describe non-adrenergic, noncholinergic (NANC) neurotransmission and the NO-cGMP pathway underlying erectile function and PDE5 inhibitor pharmacology, including the absolute nitrate contraindication.

Part 3 · BPH, Erectile Dysfunction & PDE5 Inhibitors

SO5 Integrate autonomic receptor pharmacology across the genitourinary and cardiovascular systems, and apply the full Unit 6 toolkit to integrated clinical scenarios with a cumulative recall quiz.

Part 4 · Putting It All Together: GU Toolkit

ABOUT THIS LECTURE

- **PA PHARMACOLOGY COURSE · LECTURE 6 · SESSION 2 OF 2** — Genitourinary & Reproductive Pharmacology. **Session 2: Autonomic Pharmacology of the Bladder and Lower Urinary Tract | Tuesday, December 1 | 8:30–11:00 AM** S1
- Recap: Session 1 covered gonadal steroid pharmacology (estrogens, progestins, SERMs, aromatase inhibitors, androgens/antiandrogens) and the PCOS reproductive-metabolic overlap. S2
- Today is autonomic pharmacology — mechanistically distinct from steroid hormones; the connective tissue between sessions is mostly the finasteride/dutasteride callback and the shared organ system. Three nerve supplies, three receptor families, and every drug maps to one receptor at one location. S2 (notes)
- **APPENDIX 1 — Formulation, dose, indications, adverse effects, contraindications and toxicity** — your study reference: know the dose logic, the serious adverse effects, the contraindication that changes decisions and the toxicity you would need to recognize. S43; S43 (notes); S44-57
- **PA PHARMACOLOGY COURSE · SESSION 2** · Study reference for the drugs covered in Session 2. S43
- **PO = by mouth TID = three times daily IR / ER = immediate-release / extended-release BPH = benign prostatic hyperplasia PSA = prostate-specific antigen ED = erectile dysfunction PAH = pulmonary arterial hypertension OAB = overactive bladder CNS = central nervous system** S43
- **QT = QT interval on ECG CYP = cytochrome P450 eGFR / CrCl = estimated glomerular filtration rate / creatinine clearance PDE5 = phosphodiesterase type 5 GnRH = gonadotropin-releasing hormone BP = blood pressure GI = gastrointestinal ECG = electrocardiogram NO / cGMP = nitric oxide / cyclic guanosine monophosphate** S43
- **Standard adult dosing for teaching; verify against current prescribing information.** S44-57
- **Appendix 2: DRUG PROFILE · SUPPORTING** — supporting drugs explain interactions. **Listed for context; not profiled in detail.** S58; S58 (notes)
- **Original figures in this deck:** Figures 9, 12, 13, 15 and 18 (Autonomic & Somatic Receptors of the Lower Urinary Tract; Antimuscarinics vs. Beta-3 Agonists: Two Mechanisms for Bladder Storage; Alpha1 Blockade at the Bladder Neck and Prostate; S59
- NANC Neurotransmission and the NO-cGMP Pathway in Erectile Function; Autonomic Receptor Map Across the Genitourinary Tract) were generated by the course director using the AI tool Claude — original diagrams, not reproduced from Katzung or any other copyrighted source. S59; S59 (notes)
- **Mechanism illustrations:** Figures 10, 11, 14, 16 and 17 are illustrative AI-generated schematics, not reproduced from copyrighted sources. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S59

Slide discrepancies: none found (slides, speaker notes, and study guide agree).

1 PART 1 · DRUG CLASS

Autonomic Control of the Lower Urinary Tract

SO1 IO6.1

CLASS

- **Three nerve supplies, three receptor families — the physiologic foundation for every drug in this session.** S3
- Parasympathetic M3 contracts the detrusor, sympathetic alpha-1 closes the bladder neck and beta-3 relaxes the reservoir, and somatic nicotinic control holds the external sphincter. S3 (notes); S4 (notes)
- **Figure 9. Original illustrative diagram (not from Katzung): the parasympathetic, sympathetic, and somatic nervous systems each innervate a different structure of the lower urinary tract through a distinct receptor family —** S4
- **the organizing map for every drug class covered this session.** Every class is anchored back to its receptor and anatomic location; Sections 5 and 6 "zoom in" on parts of this map. S4; S4 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Oxybutynin (blocks M3 on the detrusor), tamsulosin (blocks alpha1 at the bladder neck/prostate) and mirabegron (stimulates beta3 on the detrusor body) each target one receptor of this map. S4; S5; S7

MECHANISM OF ACTION

- **The parasympathetic pelvic nerve releases acetylcholine, which acts on M3 muscarinic receptors on the detrusor muscle to produce contraction — the primary driver of bladder emptying (voiding).** S5; SG U6-R11
- M3 is the dominant muscarinic subtype in the bladder, though M2 receptors are also present in high density (smaller functional contribution normally, may increase in disease). S5; S5 (notes)

INDICATIONS

- **Because M3 blockade reduces detrusor contraction, antimuscarinic drugs (Section 5) are a logical treatment for conditions of bladder overactivity.** S5
- **Both of these receptor subtypes become specific drug targets later in this session: alpha1 blockers for BPH (Section 6), and beta3 agonists for overactive bladder (Section 5).** S7

- Chain (Figure 10): pelvic nerve releases acetylcholine → ACh binds M3 on the detrusor → Gq activates phospholipase C; calcium rises → detrusor contracts and the bladder empties. S6; S6 (notes)
- **M3 is the dominant bladder muscarinic subtype. M2 is more abundant but contributes less to contraction. Blocking M3 underlies antimuscarinic OAB treatment.** S6
- **The sympathetic hypogastric nerve releases norepinephrine, which acts on two different adrenergic receptor subtypes in two different locations, producing two different (and complementary) effects.** S7; S7 (notes)
- Chain (Figure 11): sympathetic hypogastric nerve releases norepinephrine → alpha-1: outlet closed; beta-3: reservoir relaxed → continence maintained during storage. S8
- Alpha1 receptors at the bladder neck, trigone, and proximal urethra: norepinephrine causes contraction, helping maintain continence during storage. S7; S8; SG U6-R11
- Beta3 receptors on the detrusor body: norepinephrine causes relaxation, allowing the bladder to fill without a premature urge to void (low pressure). S7; S8; SG U6-R11

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

Not covered in materials

CONTRAINDICATIONS

Not covered in materials

DRUG INTERACTIONS

Not covered in materials

PHARMACOKINETICS

Not covered in materials

MONITORING

Not covered in materials

CLINICAL PEARL / WHY

- Net storage effect: sympathetic tone keeps the outlet closed (alpha1) while keeping the reservoir relaxed (beta3) — parasympathetic tone is low; both arms work together for continence. During voiding, this pattern reverses. S7; S8; S8 (notes)
- This is applied Chapter 6: the same muscarinic family that governs other smooth muscle and glands throughout the body — the bladder is simply one more target organ. S5; S5 (notes)
- Know which receptor does what where (alpha1 at the outlet vs. beta3 on the detrusor body) before moving on — it makes the rest of the session's pharmacology make sense. S7 (notes)

Figure 9 · Autonomic and somatic receptors of the lower urinary tract. Original illustrative diagram (not from Katzung): parasympathetic M3 (detrusor contraction), sympathetic alpha1 (bladder neck closure) and beta3 (detrusor relaxation), somatic nicotinic (external sphincter) — the map for every drug in the session. S4; S4 (notes)

Figure 10 · M3 signaling and detrusor contraction. Schematic (illustrative) — Figure 10. Acetylcholine acting on M3 receptors (Gq → phospholipase C → calcium) drives detrusor contraction and bladder emptying. S6; S6 (notes)

Figure 11 · Alpha-1 vs. beta-3: outlet closed, reservoir relaxed. Schematic (illustrative) — Figure 11. Alpha-1 contraction closes the outlet while beta-3 relaxation lets the bladder store urine — together, continence during storage. S8; S8 (notes)

Storage phase vs. voiding phase (S9)

	Storage phase	Voiding phase
Dominant tone	Sympathetic	Parasympathetic
Detrusor	Beta3 agonism relaxes the detrusor body	M3 agonism contracts the detrusor body
Bladder neck	Alpha1 agonism contracts the bladder neck/urethra	Sympathetic tone withdraws, bladder neck relaxes
External sphincter	Somatic (pudendal) tone keeps it closed	Voluntary relaxation (somatic)

How to tell them apart. Sources: S9; S9 (notes)

MNEMONIC (MEMORY AID)

- Store with the Sympathetic (alpha1 shuts the door, beta3 relaxes the tank); Pee with the Parasympathetic (M3 squeezes). S9

HIGH-YIELD

- M3 (detrusor) contracts to void; alpha1 (neck) closes and beta3 (detrusor) relaxes to store. S5; S7; SG U6-R11

2 PART 2 · DRUG CLASS

Overactive Bladder Drugs

SO2 IO6.1 IO6.2 IO6.3 IO6.4 IO6.6

CLASS

- **Two mechanisms, one goal — calming an overactive detrusor to improve storage and reduce urgency.** Antimuscarinics block the detrusor's muscarinic signal, while beta-3 agonists relax it. S12; S12 (notes)
- **Figure 13. Original illustrative diagram (not from Katzung): antimuscarinics block the M3-mediated contraction signal, while beta3 agonists enhance the norepinephrine-mediated relaxation signal —** S16
- **two distinct mechanisms converging on the same clinical goal of improved bladder storage.** Complementary, not competing — the basis for combination therapy. S16; S16 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Oxybutynin, tolterodine, solifenacin, darifenacin (antimuscarinics); mirabegron (beta3 agonist). S13; S15

MECHANISM OF ACTION

- **Mechanism: block M3 (and, to a lesser extent, M2) muscarinic receptors on the detrusor, reducing involuntary contractions and increasing functional bladder capacity.** S13; SG U6-R12
- Antimuscarinic agents differ mainly in receptor selectivity and anticholinergic side-effect burden rather than core mechanism. S13
- Chain (Figure 14): drug blocks M3 receptors on the detrusor → fewer involuntary contractions, larger capacity → same blockade in brain, salivary gland and bowel → bladder relaxes, but dry mouth, constipation and confusion follow. S14; S14 (notes)

INDICATIONS

- **Key advantage: lacks the systemic anticholinergic burden of antimuscarinics, making it a preferred option in older adults or patients already on multiple anticholinergic medications.** S15; SG U6-R13
- Mirabegron can be used alone or combined with an antimuscarinic for refractory symptoms — complementary (not redundant) mechanisms. S15; S16 (notes)

- Oxybutynin has the highest CNS anticholinergic burden; mirabegron, a beta-3 agonist, avoids this anticholinergic burden. S14
- **Mechanism: stimulate beta3 adrenergic receptors on the detrusor, enhancing smooth muscle relaxation during the storage/filling phase.** Mirabegron is the primary beta3 agonist in clinical use for overactive bladder. S15; SG U6-R13

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Adverse effects follow directly from systemic antimuscarinic blockade: dry mouth, constipation, blurred vision, and — importantly — cognitive impairment and confusion, especially in older adults.** S13; S13 (notes)
- Anticholinergic burden is cumulative across all of a patient's medications — oxybutynin has the highest CNS anticholinergic burden of this class and is often avoided in older adults or those with dementia risk. S13; S14; SG U6-R12
- Mirabegron: can raise blood pressure (monitor in uncontrolled hypertension); headache or nasopharyngitis; less commonly, urinary retention. S15

CONTRAINDICATIONS

- Antimuscarinics: caution/contraindication in narrow-angle glaucoma and significant urinary retention. S13

DRUG INTERACTIONS

- Mirabegron inhibits CYP2D6. Other anticholinergics (diphenhydramine, tricyclic antidepressants) add to anticholinergic burden; cholinesterase inhibitors (donepezil) oppose antimuscarinics. S15 (notes); S58; S58 (notes)
- Strong CYP3A4 inhibitors (ketoconazole, clarithromycin) raise solifenacin levels. S58; S58 (notes)

PHARMACOKINETICS

Not covered in materials

MONITORING

Not covered in materials

CLINICAL PEARL / WHY

- Case: a 78-year-old with mild cognitive impairment and urge incontinence starts oxybutynin; three weeks later, worsening confusion and new difficulty with daily tasks. **What is the most likely explanation, and what change in therapy would you consider?** S17
- **Oxybutynin carries a relatively high CNS anticholinergic burden, and in an older adult — especially one with pre-existing cognitive impairment — this can precipitate or worsen confusion.** S17
- **Switching to a beta3 agonist (mirabegron), which lacks anticholinergic activity, treats the same overactive bladder symptoms without this risk.** Review other anticholinergics too (medication reconciliation). S17; S17 (notes)
- The anticholinergic-burden-in-older-adults point is the highest-yield safety point in this section; "no anticholinergic burden" is mirabegron's single most important advantage. S13 (notes); S15 (notes)

Step by step

Choosing OAB therapy S13; S15; S17

Older adult, cognitive impairment, or other anticholinergics? →

Yes → mirabegron (check BP) → No → antimuscarinic (solifenacin, tolterodine)

→ Avoid with narrow-angle glaucoma or retention →

Refractory → combine both mechanisms

Figure 12 · Antimuscarinics: bladder benefit and anticholinergic burden. Schematic (illustrative) — Antimuscarinics relax the bladder but block the same receptors throughout the body (brain, salivary glands, gut): dry mouth, constipation and confusion. (The slide labels this “Figure 93”.) S14; S14 (notes)

Figure 13 · Antimuscarinics vs. beta-3 agonists. Original illustrative diagram (not from Katzung): antimuscarinics block the M3 contraction signal; beta3 agonists enhance the relaxation signal — complementary mechanisms for better storage. S16; S16 (notes)

Antimuscarinics vs. mirabegron (S13; S15)

	Antimuscarinics	Mirabegron (beta3 agonist)
Mechanism	Block M3 (and M2) on the detrusor	Stimulate beta3 on the detrusor
Adverse effects	Dry mouth, constipation, blurred vision, confusion	Raises BP; headache, nasopharyngitis
Older adults	Avoid oxybutynin (highest CNS burden)	Preferred — no anticholinergic burden
Cautions	Narrow-angle glaucoma, urinary retention	Uncontrolled hypertension; CYP2D6 inhibitor

How to tell them apart. Sources: S13; S15

Overactive bladder drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Oxybutynin (Ditropan, Oxytrol)	**Antimuscarinic (overactive bladder)**	**IR tablets 5 mg; ER tablets 5/10/15 mg; transdermal patch 3.9 mg/day; gel**	**IR: 5 mg PO two to three times daily (maximum 20 mg/day); ER: 5-10 mg daily (maximum 30 mg)**	**Overactive bladder, urge incontinence, neurogenic bladder**
Tolterodine (Detrol)	**Antimuscarinic**	**Tablets 1/2 mg; ER capsules 2/4 mg**	**IR: 2 mg PO twice daily (1 mg if hepatic impairment or strong CYP3A4 inhibitors); ER: 4 mg daily**	**Overactive bladder with urgency, frequency, and urge incontinence**
Solifenacin (Vesicare)	**Antimuscarinic (M3-selective)**	**Tablets 5/10 mg**	**5 mg PO once daily, may increase to 10 mg (maximum 5 mg with severe renal impairment or strong CYP3A4 inhibitors)**	**Overactive bladder**
Darifenacin (Enablex)	**Antimuscarinic (M3-selective)**	**ER tablets 7.5/15 mg**	**7.5 mg PO once daily, may increase to 15 mg after 2 weeks**	**Overactive bladder**
Mirabegron (Myrbetriq)	**Beta-3 adrenergic agonist**	**ER tablets 25/50 mg; granules for suspension**	**25 mg PO once daily, may increase to 50 mg after 4-8 weeks (maximum 25 mg with eGFR 15-29)**	**Overactive bladder (alone or with an antimuscarinic)**

How to tell them apart. Sources: S44; S45; S46; S47; S48

Overactive bladder drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Oxybutynin (Ditropan, Oxytrol)	**Dry mouth, constipation, blurred vision, drowsiness, confusion, urinary retention**	**Narrow-angle glaucoma, urinary retention, gastric retention, severe ulcerative colitis; avoid in older adults with dementia**	**Anticholinergic toxicity: confusion, hallucinations, tachycardia, hyperthermia, urinary retention; highest CNS burden in the class**
Tolterodine (Detrol)	**Dry mouth, constipation, headache, dizziness, dry eyes, QT prolongation (rare)**	**Urinary retention, gastric retention, uncontrolled narrow-angle glaucoma**	**Anticholinergic toxicity; reduce dose with CYP3A4 inhibitors or hepatic impairment**
Solifenacin (Vesicare)	**Dry mouth, constipation, blurred vision, QT prolongation, urinary retention**	**Urinary retention, gastric retention, uncontrolled narrow-angle glaucoma**	**Anticholinergic toxicity and QT prolongation in overdose; adjust dose with kidney, liver, and CYP3A4 interactions**
Darifenacin (Enablex)	**Constipation (prominent), dry mouth, dyspepsia, headache**	**Urinary or gastric retention, uncontrolled narrow-angle glaucoma**	**Anticholinergic toxicity; CYP2D6 and CYP3A4 interactions; limit dose with moderate CYP3A4 inhibitors**
Mirabegron (Myrbetriq)	**Hypertension, nasopharyngitis, urinary tract infection, headache, urinary retention**	**Severe uncontrolled hypertension, end-stage renal disease; caution with CYP2D6 substrates**	**Raises blood pressure; inhibits CYP2D6, increasing metoprolol and desipramine levels; no anticholinergic burden**

How to tell them apart. Sources: S44; S45; S46; S47; S48

WHY IT HAPPENS

- Why does oxybutynin confuse an older adult? It blocks muscarinic receptors in the brain too, and anticholinergic burden adds up across every drug she takes. S13; S14; S17

HIGH-YIELD

- OAB in an older adult or anyone on other anticholinergics → mirabegron; oxybutynin has the highest CNS burden. S13; S15; S17

3 PART 3 · DRUG CLASS

BPH, Erectile Dysfunction & PDE5 Inhibitors

SO3 SO4 IO6.1 IO6.2 IO6.3 IO6.4 IO6.5 IO6.6

CLASS

- **Where genitourinary pharmacology meets the cardiovascular system — alpha1 blockade, nitric oxide signaling, and a life-threatening drug interaction.** Alpha1 blockers relax the prostate; nitric oxide and cGMP drive erection. S20; S20 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Tamsulosin, silodosin, doxazosin, terazosin (alpha1 blockers); finasteride, dutasteride (5-ARIs); sildenafil, tadalafil, vardenafil (PDE5 inhibitors). S21; S23; S26

MECHANISM OF ACTION

- **Mechanism: block alpha1 receptors at the bladder neck, prostatic smooth muscle, and proximal urethra, relaxing outflow resistance and improving urinary flow in benign prostatic hyperplasia.** S21
- **Figure 14. Original illustrative diagram (not from Katzung): sympathetic norepinephrine normally contracts bladder neck and prostatic smooth muscle via alpha1 receptors, narrowing the outflow tract;** S22
- **alpha1 blockers relax this same muscle to improve urinary flow — while the same receptor family elsewhere mediates vascular tone.** S22; S22 (notes)
- **Mechanism: block 5-alpha-reductase, the enzyme that converts testosterone to the more potent dihydrotestosterone (DHT) — the same mechanism introduced in Session 1 for hirsutism and androgenetic alopecia.** S23; S23 (notes)
- 5-ARI chain (Figure 15): testosterone is converted by 5-alpha-reductase → the enzyme makes DHT → DHT drives prostate growth → enzyme blocked, DHT falls, prostate shrinks slowly. S24; S24 (notes)

INDICATIONS

- Tamsulosin, silodosin (more uroselective, less vascular effect) vs. doxazosin, terazosin (less selective, more likely to lower blood pressure systemically). S21; S21 (notes); SG U6-R14
- Finasteride, dutasteride: in BPH, reducing DHT shrinks prostatic tissue over time, improving urinary symptoms. S23
- Onset: 5-ARI benefit typically takes 3–6 months, unlike the rapid onset of alpha1 blockers — the classes are frequently combined (alpha1 blocker for rapid relief, 5-ARI for long-term disease modification). Alpha1 blockers act in days; 5-ARIs take months. S23; S24; SG U6-R15
- PDE5 agents: sildenafil, tadalafil (longer half-life, allows more flexible dosing), vardenafil. S26
- TB Finasteride is moderately effective in reducing prostate size in BPH; 1 mg/d is approved for early male pattern baldness. p. 781

- **Figure 16. Original illustrative diagram (not from Katzung): non-adrenergic, noncholinergic (NANC) nerves release nitric oxide, raising cGMP and relaxing corpus cavernosum smooth muscle;** S25
- **PDE5 normally terminates this signal by breaking down cGMP, and nitrates independently raise NO/cGMP systemically — the mechanistic basis for the absolute nitrate/PDE5-inhibitor contraindication.** S25
- **Mechanism: block phosphodiesterase type 5 (PDE5), the enzyme that normally breaks down cGMP, prolonging smooth muscle relaxation in the corpus cavernosum and sustaining erectile function (Figure 16).** S26
- Chain (Figure 17): sexual stimulation → NANC nerves release nitric oxide → guanylate cyclase makes cGMP → cGMP relaxes corpus cavernosum smooth muscle → erection. PDE5 breaks down cGMP and ends the signal. S27
- Sildenafil, tadalafil and vardenafil block PDE5 and prolong the signal; an erection still needs sexual stimulation to start the pathway. S27; S27 (notes)

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Adverse effects: orthostatic hypotension and a "first-dose effect" (pronounced hypotension with the first dose) — because alpha1 receptors also mediate vascular smooth muscle tone, not just bladder neck tone.** S21; SG U6-R14
- Floppy iris syndrome is a recognized intraoperative complication during cataract surgery in patients on alpha1 blockers — relevant to preoperative medication reconciliation. S21; S21 (notes)

CONTRAINDICATIONS

- **Absolute contraindication: concurrent use with any nitrate — both raise NO/cGMP signaling, and together they can cause profound, life-threatening hypotension.** S26
- Why absolute rather than relative: nitrates and PDE5 inhibitors independently raise NO/cGMP signaling, and together can produce profound, life-threatening hypotension — pharmacodynamic and severe regardless of dose. SG U6-R16

- **Adverse effects: decreased libido, erectile dysfunction, and — a key counseling point — these agents lower PSA levels by roughly half, which must be accounted for when interpreting PSA values in patients on long-term therapy.** S23; S24; SG U6-R15
- PDE5 common adverse effects: headache, flushing, dyspepsia, and visual disturbances (blue-tinged vision, more notable with sildenafil). S26
- **This mirrors the identical nitrate/PDE5-inhibitor warning from Unit 5's angina content, now from the opposite side of the interaction.** Riociguat is also contraindicated with PDE5 inhibitors. S26; S26 (notes); S58
- 5-ARI pregnancy caution: finasteride/dutasteride can cause ambiguous genitalia in a male fetus; pregnant partners should avoid handling crushed or broken tablets. S23; S24

DRUG INTERACTIONS

- Use caution with alpha1 blockers and PDE5 inhibitors together — additive hypotension, less severe than with nitrates, still warrants spacing doses or careful monitoring. S26; S29
- Nitrate + PDE5 inhibitor (Figure 18): nitrate raises NO and cGMP systemically → PDE5 inhibitor stops cGMP breakdown → together, vasodilation becomes profound → severe hypotension and syncope. Never combine. S29; S29 (notes)
- Antihypertensives: additive hypotension with alpha-1 blockers. Strong CYP3A4 inhibitors (ketoconazole, clarithromycin) raise PDE5 inhibitor levels. S58; S58 (notes)

PHARMACOKINETICS

- TB Finasteride lowers DHT within 8 hours, lasting about 24 hours; dutasteride has a slow onset and a much longer half-life. p. 781

MONITORING

- PSA falls about half on long-term 5-ARI therapy; read PSA accordingly. S24

CLINICAL PEARL / WHY

- **This is the exact same receptor family targeted by alpha1-blocking antihypertensives and central agents discussed in Unit 5 — the difference here is clinical intent (urinary flow vs. blood pressure), not a different drug class.** S21; S21 (notes); S22 (notes)
- Finasteride/dutasteride here are a deliberate callback to Session 1: same mechanism, same drugs, different clinical application. S23 (notes)
- Figure 16 applies the Chapter 6 non-adrenergic, noncholinergic neuron reading; the nitrate warning is the highest-stakes safety point in the unit — students saw it from the nitrate side in Unit 5, today from the PDE5 side. S25 (notes); S26 (notes)

- Case: a 64-year-old with stable angina on isosorbide mononitrate asks for an ED prescription — a friend takes sildenafil "with no problems." Ask yourself: **What should the PA do, and why?** S28
- **PDE5 inhibitors are absolutely contraindicated with any nitrate — both raise NO/cGMP signaling, and the combination can cause severe, life-threatening hypotension.** S28
- **This prescription should not be written while the patient remains on a nitrate; the conversation should include alternative angina management or alternative ED options that do not involve this interaction, in consultation with his cardiologist.** S28; S28 (notes)
- This case pulls a Unit 5 drug (isosorbide mononitrate) into a Unit 6 decision, making the shared NO-cGMP mechanism unmistakable. S28 (notes)
- Drug names reveal the class: **-osin** alpha1 blocker (tamsulosin, doxazosin, terazosin, silodosin; opens the outlet); **-terodine** antimuscarinic (tolterodine, fesoterodine; dampens detrusor M3 signal); **-fenacin** antimuscarinic (solifenacin, darifenacin; same class, same target). S32; S32 (notes)
- **oxybutynin / trospium** antimuscarinic (older, higher CNS burden); **-begron** beta-3 agonist (mirabegron, vibegron; relaxes the reservoir); **-afil** PDE5 inhibitor (sildenafil, tadalafil, vardenafil; keeps cGMP high). S32
- **-steride** 5-alpha-reductase inhibitor (finasteride, dutasteride; lowers DHT, shrinks prostate); **-nitrate / nitro-** nitrate (isosorbide, nitroglycerin; never with a PDE5 inhibitor). S32

Step by step

ED request: safe to prescribe a PDE5 inhibitor? S26; S28; S58

Any nitrate (isosorbide, nitroglycerin) or riociguat? →

Yes → do not prescribe; involve cardiology →

On an alpha1 blocker → caution, space doses →

Strong CYP3A4 inhibitor → levels rise → Counsel: needs sexual stimulation

Figure 14 · Alpha1 blockade at the bladder neck and prostate. Original illustrative diagram (not from Katzung): norepinephrine contracts bladder-neck and prostatic smooth muscle via alpha1, narrowing outflow; alpha1 blockers relax it — and the same receptors in vessels explain orthostatic hypotension. S22; S22 (notes)

Figure 15 · 5-ARIs: shrinking the prostate slowly. Schematic (illustrative) — Figure 15. 5-ARIs block DHT production, so the prostate shrinks slowly over months; combine with an alpha1 blocker for fast relief. S24; S24 (notes)

Figure 16 · NANC neurotransmission and the NO-cGMP pathway. Original illustrative diagram (not from Katzung): NANC nerves release NO → cGMP → corpus cavernosum relaxation; PDE5 breaks cGMP down; nitrates independently raise NO/cGMP — hence the absolute contraindication. S25; S25 (notes)

Figure 17 · PDE5 inhibitors: NO, cGMP and erection. Schematic (illustrative) — Figure 17. PDE5 inhibitors keep cGMP high, prolonging smooth muscle relaxation; sexual stimulation is still needed to start the pathway. S27; S27 (notes)

Figure 18 · Nitrate plus PDE5 inhibitor: a double hit on cGMP. Schematic (illustrative) — Figure 18. Nitrates and PDE5 inhibitors both raise cGMP, so vasodilation and hypotension stack — severe hypotension and syncope. Never combine; caution with alpha1 blockers too. S29; S29 (notes)

BPH drug classes (S21; S23)

	Alpha1 blockers	5-alpha-reductase inhibitors
Mechanism	Relax prostate/bladder-neck smooth muscle	Block testosterone → DHT; shrink the gland
Onset	Days	3–6 months
Adverse effects	Orthostatic hypotension, first-dose effect, floppy iris	Decreased libido, ED; PSA halved
Use	Rapid symptom relief	Long-term disease modification — often combined

How to tell them apart. Sources: S21; S23; S24

Alpha1 blocker profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Tamsulosin (Flomax)	**Alpha-1A selective blocker**	**Capsules 0.4 mg**	**0.4 mg PO once daily 30 minutes after the same meal, may increase to 0.8 mg**	**Benign prostatic hyperplasia symptoms, ureteral stone passage (off-label)**

Drug	Class	Formulation	Adult dose	Indications
Silodosin (Rapaflo)	Highly uroselective alpha-1A blocker	Capsules 4/8 mg	8 mg PO once daily with a meal (4 mg with moderate renal impairment)	Benign prostatic hyperplasia symptoms
Doxazosin (Cardura)	Alpha-1 blocker (nonselective among alpha-1 subtypes)	Tablets 1/2/4/8 mg; ER tablets 4/8 mg	BPH: 1 mg PO daily at bedtime, titrate to 4-8 mg daily; hypertension: 1-16 mg daily	Benign prostatic hyperplasia, hypertension
Terazosin (Hytrin)	Alpha-1 blocker	Tablets 1/2/5/10 mg; capsules	BPH: 1 mg PO at bedtime, titrate to 5-10 mg daily; hypertension: 1-5 mg daily (maximum 20 mg)	Benign prostatic hyperplasia, hypertension

How to tell them apart. Sources: S49; S50; S51; S52

Alpha1 blocker profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Tamsulosin (Flomax)	Dizziness, orthostatic hypotension, retrograde ejaculation, rhinitis, intraoperative floppy iris syndrome	Hypersensitivity (sulfonamide allergy caution); strong CYP3A4 inhibitors	Orthostatic hypotension and syncope with first dose; floppy iris syndrome in cataract surgery
Silodosin (Rapaflo)	Retrograde ejaculation (common), dizziness, orthostatic hypotension, diarrhea	Severe renal impairment (CrCl below 30 mL/min), severe hepatic impairment, strong CYP3A4 inhibitors	Orthostatic hypotension; floppy iris syndrome; contraindicated with strong CYP3A4 inhibitors
Doxazosin (Cardura)	Dizziness, orthostatic hypotension, first-dose syncope, fatigue, edema, floppy iris syndrome	Hypersensitivity to quinazolines; caution with PDE5 inhibitors	First-dose syncope and profound hypotension; restart at 1 mg after interruption of several days
Terazosin (Hytrin)	Dizziness, orthostatic hypotension, first-dose syncope, fatigue, peripheral edema, floppy iris syndrome	Hypersensitivity to quinazolines; caution with PDE5 inhibitors	First-dose hypotension and syncope; repeat titration after missed doses

How to tell them apart. Sources: S49; S50; S51; S52

5-alpha-reductase inhibitor profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Finasteride (Proscar, Propecia)	5-alpha-reductase inhibitor	Tablet 5 mg (BPH); 1 mg (hair loss)	5 mg PO daily for BPH; 1 mg PO daily for male-pattern hair loss	BPH, androgenetic alopecia
Dutasteride (Avodart)	5-alpha-reductase inhibitor (types 1 and 2)	Capsules 0.5 mg; combination with tamsulosin (Jalyn)	0.5 mg PO once daily for BPH; effects take 3-6 months	Benign prostatic hyperplasia, combination therapy with an alpha-1 blocker

How to tell them apart. Sources: S53; S54

5-alpha-reductase inhibitor profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Finasteride (Proscar, Propecia)	Decreased libido, erectile dysfunction, reduced ejaculate, gynecomastia	Pregnancy and women who may be pregnant (do not handle crushed tablets), hypersensitivity	Lowers PSA by about half; reports of depression; teratogenic to a male fetus
Dutasteride (Avodart)	Decreased libido, erectile dysfunction, ejaculation disorders, breast tenderness, depression	Pregnancy and women of childbearing potential handling capsules, severe hepatic impairment	Lowers PSA by about half; pregnant partners must avoid handling capsules; monitor PSA

How to tell them apart. Sources: S53; S54

PDE5 inhibitor profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Sildenafil (Viagra, Revatio)	PDE5 inhibitor	Tablets 25/50/100 mg (Viagra); 20 mg (Revatio)	ED: 50 mg PO about 1 hour before activity (range 25-100 mg), no more than once daily; PAH: 20 mg TID	Erectile dysfunction, pulmonary arterial hypertension (Revatio)
Tadalafil (Cialis, Adcirca)	PDE5 inhibitor (long-acting)	Tablets 2.5/5/10/20 mg	ED: 10 mg PO before activity (5-20 mg), or 2.5-5 mg daily; BPH: 5 mg daily	Erectile dysfunction, BPH symptoms, pulmonary arterial hypertension (Adcirca)
Vardenafil (Levitra, Staxyn)	PDE5 inhibitor	Tablets 2.5/5/10/20 mg; orally disintegrating tablet 10 mg	10 mg PO about 60 minutes before activity (5-20 mg), no more than once daily	Erectile dysfunction

How to tell them apart. Sources: S55; S56; S57

PDE5 inhibitor profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Sildenafil (Viagra, Revatio)	**Headache, flushing, dyspepsia, nasal congestion, visual changes (blue tint), hypotension**	**Concurrent nitrates or riociguat, hypersensitivity; caution with alpha-1 blockers and severe cardiovascular disease**	**Profound hypotension with nitrates; priapism over 4 hours is an emergency; sudden vision or hearing loss**
Tadalafil (Cialis, Adcirca)	**Headache, dyspepsia, back pain, myalgia, flushing, nasal congestion**	**Concurrent nitrates or riociguat; caution with alpha-1 blockers (except tamsulosin 0.4 mg) and strong CYP3A4 inhibitors**	**Profound hypotension with nitrates for at least 48 hours after the last dose; priapism**
Vardenafil (Levitra, Staxyn)	**Headache, flushing, rhinitis, dyspepsia, QT prolongation, visual changes**	**Concurrent nitrates or riociguat, congenital long QT, class IA or III antiarrhythmics**	**Profound hypotension with nitrates; QT prolongation; priapism**

How to tell them apart. Sources: S55; S56; S57

WHY IT HAPPENS

- Why does tamsulosin cause dizziness on standing? Alpha1 receptors in blood vessels are blocked too; the uroselective drugs do it less. S21; S42

HIGH-YIELD

- BPH: alpha1 blocker (days; orthostasis, first dose) + 5-ARI (3–6 months; PSA halved) — often together. S21; S23
- PDE5 inhibitor + any nitrate = absolute contraindication; caution with alpha1 blockers. S26; S29; SG U6-R16

4 PART 4 · TOPIC

Putting It All Together: GU Toolkit

SO5 IO6.3 IO6.4 IO6.6

KEY DEFINITIONS

- **Applying the whole genitourinary and reproductive pharmacology toolkit, across both sessions.** S33
- **Figure 19. Original illustrative diagram (not from Katzung): the detrusor, bladder neck/prostate, and corpus cavernosum each rely on receptor families — muscarinic, adrenergic, and NO-cGMP signaling — that recur in the systemic vasculature,** S34
- **explaining why so many genitourinary drug classes have direct cardiovascular counterparts or interactions.** Receptor families are shared with the vasculature — the unifying theme of the session. S34; S34 (notes)

HOW IT WORKS

- Session 1's gonadal hormone pharmacology (Ch. 40) and PCOS (Ch. 41 overlap) and Session 2's autonomic bladder pharmacology (Ch. 6) are distinct but connect through shared classes — 5-alpha-reductase inhibitors appear in both sessions. S38
- The lower urinary tract is governed by three nerve supplies and three receptor families (Figure 9): parasympathetic/M3, sympathetic/alpha1 and beta3, and somatic/nicotinic. S38
- Antimuscarinics and beta3 agonists treat overactive bladder through complementary mechanisms — blocking contraction vs. enhancing relaxation — with anticholinergic burden the key safety differentiator in older adults. S38
- Alpha1 blockers and 5-ARIs treat BPH through different timescales and mechanisms and are often combined; alpha1 blockade overlaps directly with Unit 5's antihypertensive pharmacology. S38
- The NO-cGMP pathway (Figure 16) underlies both erectile function and nitrate angina therapy (Unit 5) — therapeutic in one context, dangerous in combination in another: one of the highest-stakes safety points in the course. S38
- Close by naming the cross-unit integration: genitourinary pharmacology shares receptor families and drug classes with cardiovascular pharmacology covered earlier in the course. S38 (notes)

DRUGS NAMED AS EXAMPLES

- **A cumulative self-test spanning both sessions — use this to review before exams.** S39
- **Recall Q1:** unopposed estrogen drives unchecked endometrial proliferation, raising hyperplasia and cancer risk with a uterus — a progestin is always paired unless she has had a hysterectomy. S40; S42
- **Recall Q2:** tamoxifen is an endometrial agonist (cancer risk) and breast antagonist; raloxifene shares breast antagonist and bone agonist effects but is essentially neutral at the endometrium. S40; S42
- **Recall Q3:** hyperinsulinemia stimulates ovarian theca-cell androgen production and lowers hepatic SHBG, raising free testosterone — the center of the PCOS cycle (Figure 7). S40; S42
- **Recall Q4:** a patient on an antimuscarinic for overactive bladder develops new confusion — switch to mirabegron, which enhances beta3-mediated detrusor relaxation without blocking muscarinic receptors. S41; S42
- That avoids the cumulative anticholinergic burden that can precipitate or worsen confusion in susceptible patients, particularly older adults. S42
- **Recall Q5:** alpha1 receptors are not bladder-specific — blockade also lowers systemic vascular resistance (orthostatic hypotension, first-dose effect). **Q6:** nitrates and PDE5 inhibitors both raise NO/cGMP; combined, potentially fatal hypotension. S41; S42

CLINICAL

- **Patient A — older adult with urge incontinence and mild cognitive impairment, needs therapy that avoids anticholinergic burden:** mirabegron (Section 5). S35; S35 (notes)
- **Patient B — man with BPH, wants both rapid symptom relief and long-term disease modification:** combine an alpha1 blocker with a 5-alpha-reductase inhibitor (Section 6). S35; S35 (notes)

- **Patient C — woman with PCOS, irregular cycles, and hirsutism, not currently trying to conceive:** combined hormonal contraception ± metformin (Session 1). S35; S35 (notes)
- **Patient D — man with stable angina on a long-acting nitrate, requesting treatment for erectile dysfunction:** PDE5 inhibitors are contraindicated; this requires a cardiology-involved conversation about alternatives. S35; S35 (notes)

TRAP

- Integrated case: a PDE5 inhibitor should not be prescribed while this patient remains on isosorbide mononitrate — both drugs independently raise NO/cGMP signaling; the combination can cause profound, life-threatening hypotension. SG U6-S
- This is an absolute contraindication, and any ED treatment decision must involve his cardiologist and consideration of whether his anginal nitrate therapy can be modified first. SG U6-S
- Even without the nitrate, tamsulosin and a PDE5 inhibitor both lower blood pressure (alpha1 blockade and NO-cGMP potentiation) — additive orthostatic hypotension warrants spacing doses or close monitoring. SG U6-S
- His wife's new vaginal bleeding on tamoxifen should not be dismissed — tamoxifen's endometrial agonism raises hyperplasia/cancer risk; she needs prompt gynecologic evaluation. SG U6-S
- Match each drug or drug class to its mechanism or site of action — tamoxifen: breast antagonist, bone/endometrium agonist; raloxifene — endometrium neutral; finasteride/dutasteride — 5-alpha-reductase inhibition; oxybutynin — M3 (and M2) antagonism on the detrusor. SG U6-M
- Match: mirabegron — beta3 agonism on the detrusor; tamsulosin — uroselective alpha1 antagonism; sildenafil/tadalafil — PDE5 inhibition prolonging cGMP; metformin (PCOS) — insulin sensitization, indirectly lowering ovarian androgens. SG U6-M

Figure 19 · Autonomic receptor map across the genitourinary tract. Original illustrative diagram (not from Katzung): the detrusor (muscarinic), bladder neck/prostate (adrenergic) and corpus cavernosum (NO-cGMP) use receptor families that recur in the systemic vasculature — why GU drugs have cardiovascular counterparts and interactions. S34; S34 (notes)

Which approach would you choose? (S35)

Patient	Approach	Biggest safety check
A · Older adult, urge incontinence, mild cognitive impairment	Mirabegron	Blood pressure; avoid anticholinergics
B · BPH, wants fast relief and disease modification	Alpha1 blocker + 5-ARI	Orthostasis, first dose; PSA halved
C · PCOS, irregular cycles, hirsutism, no pregnancy plans	Combined contraception ± metformin	Estrogen contraindications
D · Angina on a long-acting nitrate, wants ED treatment	No PDE5 inhibitor; involve cardiology	Life-threatening hypotension

How to tell them apart. Sources: S35; S35 (notes)

Supporting drugs at a glance (S58)

Drug(s)	Why it matters here
Nitrates	Isosorbide, nitroglycerin: never combine with PDE5 inhibitors
Riociguat	Contraindicated with PDE5 inhibitors

Drug(s)	Why it matters here
Other anticholinergics	Diphenhydramine, tricyclic antidepressants: add to anticholinergic burden
Cholinesterase inhibitors	Donepezil: oppose antimuscarinics
Antihypertensives	Additive hypotension with alpha-1 blockers
Strong CYP3A4 inhibitors	Ketoconazole, clarithromycin: raise PDE5 inhibitor and solifenacin levels
Desmopressin	Nocturia, beyond this lecture
OnabotulinumtoxinA	Refractory overactive bladder, beyond this lecture
Duloxetine	Stress incontinence, beyond this lecture

How to tell them apart. Sources: S58; S58 (notes)

MNEMONIC (MEMORY AID)

- Every case, two answers: **the approach** and **the one thing to check**. S35 (instructor script)

HIGH-YIELD

- GU receptor families recur in the vasculature: alpha1 blockers lower BP; PDE5 inhibitors and nitrates share cGMP. S34; S38

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Detrusor contraction	M3 (Gq → PLC → Ca^{2+}) S5; S6
Storage receptors	Alpha1 (neck closed) + beta3 (detrusor relaxed) S7; S8
Antimuscarinic AEs	Dry mouth, constipation, blurred vision, confusion S13
Highest CNS burden	Oxybutynin S13; SG U6-R12
Mirabegron	Beta3 agonist; no anticholinergic burden; ↑ BP; CYP2D6 inhibitor S15
Uroselective alpha1	Tamsulosin, silodosin S21; SG U6-R14
Alpha1 blocker AEs	Orthostatic hypotension, first-dose effect, floppy iris S21
5-ARI	3–6 months; PSA halved; pregnant partners avoid crushed tablets S23; SG U6-R15
PDE5 + nitrate	Absolute contraindication — profound hypotension S26; SG U6-R16
Tadalafil	Longer half-life S26
Sildenafil	Blue-tinged vision S26
Finasteride onset	DHT falls within 8 h; prostate shrinks over months p. 781; S23

Self-test

First, the lecture's own 10 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S10-11

Which receptor and location combination is primarily responsible for keeping the bladder outlet closed during the storage phase?

- **A** Alpha1 receptors at the bladder neck and proximal urethra
- **B** M3 receptors on the detrusor muscle
- **C** Beta3 receptors on the detrusor muscle
- **D** Nicotinic receptors at the detrusor muscle

ANSWER: A

Sympathetic norepinephrine acting on alpha1 receptors at the bladder neck and proximal urethra produces smooth muscle contraction, keeping the outlet closed during storage. Beta3 receptors, by contrast, relax the detrusor body (not the outlet) during this same phase. S10-11

Lecture check 2 · S18-19

Which overactive bladder medication is generally preferred in an older adult already taking multiple anticholinergic medications?

- **A** Mirabegron, a beta3 agonist with no anticholinergic activity
- **B** Oxybutynin, for its high potency
- **C** Any antimuscarinic agent, since they are interchangeable
- **D** No pharmacologic therapy should be used in older adults

ANSWER: A

Because mirabegron works by enhancing beta3-mediated detrusor relaxation rather than blocking muscarinic receptors, it avoids adding to a patient's cumulative anticholinergic burden — a key consideration in older adults, who are more susceptible to anticholinergic-related confusion and cognitive impairment. S18-19

Lecture check 3 · S30-31

Why is the combination of a nitrate and a PDE5 inhibitor absolutely contraindicated?

- **A** Both drugs independently raise NO/cGMP signaling, and together can cause life-threatening hypotension
- **B** Nitrates block the PDE5 enzyme, making PDE5 inhibitors ineffective
- **C** The combination has no significant interaction and is commonly prescribed together
- **D** PDE5 inhibitors increase nitrate metabolism, reducing anti-anginal efficacy

ANSWER: A

Nitrates raise NO/cGMP systemically (the mechanism behind their anti-anginal effect), while PDE5 inhibitors prevent cGMP breakdown. Combined, these two independent mechanisms produce profound vasodilation and can precipitate severe, potentially fatal hypotension — the same NO-cGMP pathway shown in Figure 16, now viewed as a drug-interaction risk rather than a therapeutic mechanism. S30-31

Lecture check 4 · S36-37

Across this unit, what is the common thread linking alpha1 blockers' orthostatic hypotension risk and the nitrate/PDE5-inhibitor contraindication?

- **A** Both reflect genitourinary drug classes acting on receptor families (adrenergic, NO-cGMP) that are shared with the systemic vasculature
- **B** Both are reasons to avoid all genitourinary pharmacology in older adults
- **C** Both are unrelated adverse effects with no shared mechanism
- **D** Both apply only to patients with no cardiovascular disease

ANSWER: A

Alpha1 blockers act on the same receptor family in the bladder neck/prostate and in blood vessels, and PDE5 inhibitors/nitrates both converge on the same NO-cGMP signaling pathway used in both the corpus cavernosum and the vasculature. Recognizing receptor families as the shared variable — not organ system — is what ties genitourinary and cardiovascular pharmacology together across Units 5 and 6. S36-37

Lecture check 5 · S40; S42

Why is unopposed estrogen avoided in a patient with an intact uterus?

- **A** It significantly increases the risk of endometrial hyperplasia and cancer
- **B** It has no effect on the endometrium
- **C** It always causes immediate heavy bleeding
- **D** It is only a concern in postmenopausal patients

ANSWER: A

Unopposed estrogen drives unchecked endometrial proliferation, significantly raising the risk of endometrial hyperplasia and cancer in a patient with a uterus — this is why a progestin is always paired with estrogen therapy unless the patient has had a hysterectomy. S40; S42

Lecture check 6 · S40; S42

Why does tamoxifen carry an increased risk of endometrial cancer while raloxifene largely does not?

- **A** Tamoxifen acts as an estrogen receptor agonist in the endometrium; raloxifene is essentially neutral there
- **B** Tamoxifen blocks the androgen receptor instead of the estrogen receptor
- **C** Raloxifene has no effect on the estrogen receptor anywhere in the body
- **D** Tamoxifen and raloxifene have identical endometrial effects

ANSWER: A

Tamoxifen is a tissue-selective estrogen receptor modulator that acts as an agonist in the endometrium (raising cancer risk) while acting as an antagonist in breast tissue; raloxifene shares the breast antagonist and bone agonist effects but is essentially neutral at the endometrium. S40; S42

Lecture check 7 · S40; S42

What is the central pathophysiologic driver linking insulin resistance to hyperandrogenism in PCOS?

- **A** Hyperinsulinemia stimulates ovarian androgen production and lowers hepatic SHBG, raising free testosterone
- **B** Insulin resistance has no mechanistic connection to androgen levels
- **C** Hyperandrogenism always precedes and causes insulin resistance, never the reverse
- **D** Metformin directly blocks the androgen receptor

ANSWER: A

Hyperinsulinemia stimulates ovarian theca-cell androgen production directly and lowers hepatic SHBG production, raising free testosterone — the mechanistic link at the center of the PCOS-insulin resistance cycle (Figure 7). S40; S42

Lecture check 8 · S41; S42

A patient on an antimuscarinic for overactive bladder develops new confusion. What is the most appropriate next step?

- **A** Switch to a beta3 agonist (mirabegron), which lacks anticholinergic activity
- **B** Increase the antimuscarinic dose
- **C** Add a second antimuscarinic agent
- **D** Confusion is unrelated to antimuscarinic therapy

ANSWER: A

Mirabegron enhances beta3-mediated detrusor relaxation without blocking muscarinic receptors, avoiding the cumulative anticholinergic burden that can precipitate or worsen confusion in susceptible patients, particularly older adults. S41; S42

Lecture check 9 · S41; S42

Why are alpha1 blockers used for BPH associated with orthostatic hypotension?

- **A** Alpha1 receptors also mediate vascular smooth muscle tone, not just bladder neck tone
- **B** Alpha1 blockers have no effect outside the bladder
- **C** Orthostatic hypotension only occurs with beta3 agonists
- **D** Alpha1 blockers directly block cardiac beta receptors

ANSWER: A

Alpha1 receptors are not bladder-specific — the same receptor family mediates vascular smooth muscle tone throughout the body, so alpha1 blockade at the bladder neck also lowers systemic vascular resistance, producing orthostatic hypotension and a first-dose effect. S41; S42

Lecture check 10 · S41; S42

Why is the combination of a nitrate and a PDE5 inhibitor absolutely contraindicated?

- **A** Both independently raise NO/cGMP signaling, and together can cause life-threatening hypotension
- **B** PDE5 inhibitors inactivate nitrates before they reach the bloodstream
- **C** The combination has no significant interaction
- **D** Nitrates block PDE5 directly, making the combination redundant but safe

ANSWER: A

Nitrates and PDE5 inhibitors both raise NO/cGMP signaling through independent mechanisms; combined, the effect on vascular smooth muscle relaxation is additive to synergistic, producing profound and potentially fatal hypotension. S41; S42

Original questions

Q1 · Mechanism · Autonomic Control of the Lower Urinary Tract

Which receptor on the detrusor body relaxes the bladder during filling?

- **A** M3 muscarinic receptor
- **B** Beta3 adrenergic receptor
- **C** Nicotinic receptor
- **D** Alpha1 adrenergic receptor
- **E** M2 muscarinic receptor

ANSWER: B

Norepinephrine at beta3 on the detrusor body causes relaxation during storage (S7). S7; S8; SG U6-R11

- **A.** Contracts the detrusor (S5).
- **C.** External sphincter (S9).
- **D.** Contracts the bladder neck (S7).
- **E.** Smaller contraction role (S5).

Q2 · Mechanism · Autonomic Control of the Lower Urinary Tract

During the storage phase, which pattern of autonomic tone is correct?

- **A** Both systems are silent during storage
- **B** Sympathetic contracts the detrusor body
- **C** Sympathetic dominates; outlet closed
- **D** Somatic tone opens the external sphincter
- **E** Parasympathetic dominates; detrusor contracts

ANSWER: C

Sympathetic tone keeps the outlet closed (alpha1) and reservoir relaxed (beta3); parasympathetic tone is low (S7; S9). S7; S9

- **A.** Sympathetic dominates (S9).
- **B.** Beta3 relaxes it (S7).
- **D.** It keeps it closed (S9).
- **E.** That is voiding (S9).

Q3 · Mechanism · Overactive Bladder Drugs

How does mirabegron improve overactive bladder?

- **A** Blocks M3 on the detrusor
- **B** Stimulates beta3 to relax the detrusor
- **C** Blocks alpha1 at the bladder neck
- **D** Raises cGMP in smooth muscle
- **E** Lowers DHT to shrink the prostate

ANSWER: B

Mirabegron enhances beta3-mediated detrusor relaxation without blocking muscarinic receptors (S15; S42). S15; S42

- **A.** That is antimuscarinics (S13).
- **C.** That is tamsulosin (S21).
- **D.** That is PDE5 inhibitors (S26).
- **E.** That is finasteride (S23).

Q4 · Clinical vignette · Overactive Bladder Drugs

A patient with uncontrolled hypertension needs OAB therapy. What concern applies to mirabegron?

- **A** It causes severe confusion
- **B** It lowers PSA by half
- **C** It worsens narrow-angle glaucoma
- **D** It can raise blood pressure
- **E** It causes first-dose hypotension

ANSWER: D

Mirabegron can raise blood pressure; monitor in uncontrolled hypertension (S15; S48). S15; S48

- **A.** Antimuscarinics (S13).
- **B.** 5-ARIs (S23).
- **C.** Antimuscarinics (S13).
- **E.** Alpha1 blockers (S21).

Q5 · Clinical vignette · BPH, Erectile Dysfunction & PDE5 Inhibitors

A man with BPH wants fast relief and long-term shrinkage of his prostate. Best plan?

- **A** Oxybutynin for outflow obstruction
- **B** Doxazosin alone for shrinkage
- **C** Sildenafil plus a nitrate
- **D** 5-ARI alone for fast relief
- **E** Alpha1 blocker plus a 5-ARI

ANSWER: E

Alpha1 blockers act in days; 5-ARIs take 3–6 months to modify disease — combine them (S23–S24). S23; S24; S35

- **A.** Risks retention (S13).
- **B.** No shrinkage (S21).
- **C.** Contraindicated (S26).
- **D.** Takes months (S23).

Q6 · Clinical vignette · BPH, Erectile Dysfunction & PDE5 Inhibitors

A man on finasteride for 2 years has a PSA of 1.8 ng/mL. How should it be read?

- **A** As meaningless on therapy
- **B** At face value, 1.8 ng/mL
- **C** As about 0.9 ng/mL
- **D** As about 3.6 ng/mL (double it)
- **E** Only after a 6-month drug holiday

ANSWER: D

5-ARIs lower PSA by roughly half on long-term therapy (S23–S24). S23; S24; SG U6-R15

- **A.** Still usable (S23).
- **B.** It is halved (S23).
- **C.** Doubled, not halved (S23).
- **E.** Not the taught approach (S23).

Q7 · Clinical vignette · BPH, Erectile Dysfunction & PDE5 Inhibitors

A patient starting doxazosin for BPH is told to take the first dose at bedtime. Why?

- **A** To lower PSA faster
- **B** To prevent nighttime urinary retention
- **C** To avoid first-dose hypotension
- **D** To avoid anticholinergic confusion
- **E** Because it works only at night

ANSWER: C

Alpha1 blockers cause a first-dose effect and orthostatic hypotension (S21). S21; S51

- **A.** 5-ARIs (S23).
- **B.** Not the reason (S21).
- **D.** Antimuscarinics (S13).
- **E.** Not the reason (S21).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO6.1** Identify major pharmacologic classes used in the management of genitourinary and reproductive conditions and describe their mechanisms of action.
- [ ] **IO6.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat genitourinary and reproductive conditions.
- [ ] **IO6.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO6.4** Select appropriate pharmacologic therapy for common genitourinary and reproductive conditions based on clinical presentation and patient-specific factors.
- [ ] **IO6.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in genitourinary and reproductive conditions.
- [ ] **IO6.6** Explain important medication-use and safety information to patients receiving genitourinary and reproductive pharmacotherapy.
- [ ] **SO1** Describe the autonomic and somatic innervation of the bladder and lower urinary tract, and the receptor subtypes governing storage versus voiding.

- [ ] **SO2** Describe the mechanism, indications, and adverse effects of antimuscarinic agents and beta-3 agonists in overactive bladder and urinary incontinence.
- [ ] **SO3** Describe the mechanism and adverse effects of alpha1 blockers and 5-alpha-reductase inhibitors in benign prostatic hyperplasia (BPH).
- [ ] **SO4** Describe non-adrenergic, noncholinergic (NANC) neurotransmission and the NO-cGMP pathway underlying erectile function and PDE5 inhibitor pharmacology, including the absolute nitrate contraindication.
- [ ] **SO5** Integrate autonomic receptor pharmacology across the genitourinary and cardiovascular systems, and apply the full Unit 6 toolkit to integrated clinical scenarios with a cumulative recall quiz.